

Shared Causal Convolutions with Lexical Residual Objects for Attention-Free Language Modeling

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Abstract

We study whether a small causal language model can separate sequence transport from token-specific computation without query-key-value attention. The proposed model uses a shared stack of dilated depthwise causal convolutions for sequence mixing, a tied low-rank object indexed by token identity, and narrow deterministic micro-expert residuals. The resulting decode state has fixed size with respect to generated context length and requires no attention matrix or growing key-value cache. We specify matched controls that isolate the sequence mixer, lexical object, and micro-expert residual at approximately 100M parameters. This draft intentionally withholds headline numerical claims until the preregistered one-billion-token runs and held-out evaluations are complete. The study is designed to report both matched-token language-model quality and hardware-specific training and incremental-decode throughput.

1 Introduction

Attention-based language models couple two functions: transporting information across positions and dynamically selecting which positions interact. This coupling is expressive, but autoregressive deployment typically stores a key-value (KV) cache whose size grows with context length. We ask a narrower empirical question: at the 100M-parameter scale, can a regular causal sequence substrate carry context while small token-conditioned residuals provide specialization?

Our design follows a shared-plus-residual principle. Every token traverses the same dense, causal computation path. Token identity only modulates narrow residual branches. The shared sequence mixer is a stack of depthwise causal convolutions with cycling dilations; it contains no query, key, value, attention softmax, or positional attention operation. A tied low-rank lexical table and deterministic micro-experts then adjust the shared transformation.

The intended contributions are empirical rather than claims that any individual component is new:

- a controlled integration of a shared causal-convolution substrate with tied lexical residual objects and narrow deterministic micro-experts;
- matched-parameter ablations that separate the effects of sequence mixing and lexical specialization;
- an exact incremental-decode formulation with fixed-size state, tested for equivalence to full-sequence evaluation; and
- a transparent account of quality, training throughput, decode throughput, memory, and failed ablations at the 100M scale.

We deliberately avoid claiming a universal replacement for attention. The study tests a specific scale, tokenizer, corpus, and receptive-field schedule.

2 Method

2.1 Shared causal sequence mixer

Let $\mathbf{H} \in \mathbb{R}^{B \times T \times d}$ be normalized hidden states. Layer ℓ uses a depthwise causal convolution with kernel width k and dilation

$$\delta_\ell = 2^{\ell \bmod C}, \quad (1)$$

where C is the dilation cycle. After left padding by $(k-1)\delta_\ell$, the mixer is

$$\mathbf{C}_\ell = \text{DWConv}_{k, \delta_\ell}(\mathbf{H}), \quad (2)$$

$$\mathbf{M}_\ell = \mathbf{W}_o [\text{SiLU}(\mathbf{W}_g \mathbf{C}_\ell) \odot (\mathbf{W}_u \mathbf{C}_\ell)]. \quad (3)$$

All tokens use the same parameters and computation. No QKV projection or attention matrix is instantiated. For incremental decoding, layer ℓ stores only its last $(k-1)\delta_\ell$ input vectors. The total state per sequence is

$$S = d \sum_{\ell=0}^{L-1} (k-1)\delta_\ell, \quad (4)$$

which is independent of the number of previously generated tokens. The implementation updates preallocated state buffers in place under inference mode.

2.2 Tied lexical residual object

The shared feed-forward transformation is a dense SwiGLU,

$$\mathbf{F}(\mathbf{h}) = \mathbf{W}_d [\text{SiLU}(\mathbf{W}_g \mathbf{h}) \odot \mathbf{W}_u \mathbf{h}]. \quad (5)$$

For token identity x_t , a tied table $\mathbf{E} \in \mathbb{R}^{|\mathcal{V}| \times r}$ provides a low-rank modulation:

$$\mathbf{z}_t = \text{SiLU}(\mathbf{U} \mathbf{h}_t) \odot (\mathbf{1} + \mathbf{E}[x_t]), \quad (6)$$

$$\mathbf{O}(\mathbf{h}_t, x_t) = \mathbf{D} \mathbf{z}_t. \quad (7)$$

The same table $\mathbf{E}$ is shared across layers; projection matrices remain layer-specific. The branch is initialized neutrally by setting $\mathbf{E}$ to zero and using a small learned output gate α :

$$\mathbf{F}_{\text{object}}(\mathbf{h}_t, x_t) = \mathbf{F}(\mathbf{h}_t) + \alpha \mathbf{O}(\mathbf{h}_t, x_t). \quad (8)$$

2.3 Deterministic micro-expert residual

Each layer assigns token x_t to one of N micro-experts using

$$e_\ell(x_t) = (x_t + \ell) \bmod N. \quad (9)$$

All expert hidden activations are computed as one regular tensor operation and the selected output is gathered without sorting or scatter. With expert width w ,

$$\mathbf{R}_\ell(\mathbf{h}_t, x_t) = \beta \mathbf{W}_d^{(e_\ell(x_t))} \left[\text{SiLU}(\mathbf{W}_g^{(e_\ell(x_t))} \mathbf{h}_t) \odot \mathbf{W}_u^{(e_\ell(x_t))} \mathbf{h}_t \right]. \quad (10)$$

The complete feed-forward output is $\mathbf{F} + \alpha \mathbf{O} + \beta \mathbf{R}$. This is a lexical residual over a shared dense path, not a claim that routing replaces shared computation.

Table 1: Headline matched-token results. Cells remain pending until each run reaches its registered token budget and its metrics artifact is archived.

Model	Parameters	Held-out loss ↓	Train tok/s ↑
Dense GQA baseline	pending	PENDING	pending
Causal convolution, dense FFN	pending	pending	pending
Causal convolution + lexical object	pending	pending	pending
Full lexical micro-expert model	98,194,244	PENDING	PENDING

3 Experimental Design

3.1 Matched controls

The headline study compares four approximately 100M-parameter causal language models:

1. a dense grouped-query-attention Transformer with dense SwiGLU;
2. causal convolution with dense SwiGLU;
3. causal convolution with the tied lexical object but no micro-expert residual; and
4. the complete causal convolution, tied lexical object, and micro-expert model.

Parameter counts, tokenizer, corpus, held-out partition, optimizer, learning-rate schedule, precision, sequence length, effective token batch, random seed, and total training tokens are held fixed or reported explicitly where exact equality is impossible.

3.2 Data and tokenizer

Models use the 200,019-entry `o200k_base` tokenizer and FineWeb-Edu sample-10BT. Documents with source-stream index congruent to zero modulo 20 form a fixed held-out stream and are excluded from training. The remaining 95% form the training stream. The headline budget is 1,000,013,824 training tokens per condition at sequence length 2,048.

3.3 Metrics

We report held-out next-token cross-entropy, parameter count, realized token count, training throughput, peak accelerator memory, prefill throughput, and batch-one incremental-decode latency. Matched-token quality and wall-clock efficiency are treated as separate claims. Decode correctness requires cached token-by-token logits to match full-sequence logits within stated numerical tolerance.

3.4 Hardware

Headline training runs use a single NVIDIA H100 80GB HBM3 in BF16. Apple M5 runs are reported only as short diagnostics and are not used as headline quality evidence.

4 Results

4.1 Diagnostic evidence

Before the headline runs, short matched-parameter diagnostics were used only to reject broken configurations and select the candidate. On Apple M5, the full 98.2M attention-free candidate reached held-out loss 7.557967 after 204,800 tokens, versus 7.944988 for a saved dense control, while the last measured training throughput

was 6,145 versus 4,779 tokens/s. These short-run values are not treated as final evidence because early optimization dynamics may differ across architectures.

A persistent diagonal recurrent-state extension was also tested. It was numerically stabilized with chunkwise parallel evaluation, but underperformed the simpler convolutional candidate in both loss and throughput at sequence lengths 256 and 1,024. We therefore retain the simpler model as the registered candidate rather than adding an unsupported mechanism.

4.2 Incremental equivalence and fixed state

For the selected model, incremental cached decoding reproduces full-sequence logits at absolute and relative tolerance 10^{-5} in unit tests. Cache addresses remain stable over repeated decode calls. The ten-layer, width-384 configuration stores 297,216 state elements per sequence, or 580.5 KiB in BF16/FP16, independent of generated context length. CUDA Graph capture and target-device decode benchmarks remain pending and will not be claimed until executed.

5 Related Work

Causal convolutions, state-space sequence models, recurrent attention alternatives, conditional computation, and lexical memory mechanisms all motivate parts of this study. We do not claim novelty for depthwise causal convolution, deterministic expert assignment, shared expert computation, or token-indexed embeddings individually. The question studied here is whether their constrained integration provides a useful quality-throughput-state tradeoff at small language-model scale. This section will be completed from a verified closest-prior-work review before submission; citations are intentionally not guessed in the initial draft.

6 Limitations and Broader Impact

The current study is limited to approximately 100M parameters, one primary tokenizer, one web-text corpus, and a finite set of seeds. A favorable result would not establish scaling behavior or universal superiority to attention. The convolutional receptive field is finite; fixed decode state is obtained partly by declining unbounded access to the complete prefix. Hardware throughput is implementation- and device-specific and must not be generalized from Apple M5 or a single H100.

Token-identity conditioning may encode corpus-specific frequency and lexical biases. FineWeb-Edu inherits limitations of large web corpora, including uncertain provenance, representational imbalance, and potentially harmful content. The proposed architecture does not remove these risks. Its fixed-state inference could reduce serving memory and energy, but efficiency gains can also lower the cost of deploying low-quality or harmful models. We report data provenance, held-out construction, and deployment constraints so these tradeoffs can be assessed.

7 Conclusion

We formulate attention-free language modeling as a separation between shared causal transport and narrow lexical specialization. The resulting candidate uses regular tensor operations and fixed-size incremental state. Final conclusions are deferred until all matched one-billion-token controls and held-out evaluations are complete. If the candidate does not outperform its controls, the negative result will still identify which integration choices fail under a matched budget and will narrow the claims accordingly.

References

A Reproducibility Checklist

The final anonymous supplement will include:

- the exact model source and realized configuration for every reported condition;
- tokenizer name, vocabulary size, dataset revision, and deterministic train/eval partition rule;
- parameter counts and exact token-budget assertions;
- raw per-step CSV metrics and scripts that regenerate tables;
- environment, accelerator, CUDA, and PyTorch versions;
- tests for causality, absence of QKV parameters, tied-object identity, full/incremental equivalence, and fixed cache addresses; and
- commands for training and inference benchmarks without credentials, hostnames, local paths, or repository history.

The review artifact will exclude checkpoints unless required for a reported evaluation, as well as Git history, author metadata, cloud addresses, and transient logs.